

1 Estimating the SLR Parameters

Introduction

- OLS: choose β_0, β_1 to minimize $\sum_i (y_i - \beta_0 - \beta_1 x_i)^2$
- MLE: given a full distributional assumption on the errors, choose parameter values that make the observed data most probable

The Likelihood Function

Since $Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$, and due to the observations being independent:

$$\begin{aligned} L(\beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \\ \ell(\beta_0, \beta_1, \sigma^2) &= -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_i (y_i - \beta_0 - \beta_1 x_i)^2 \end{aligned} \quad (1)$$

Maximizing Over β_0 and β_1

- For fixed σ^2 , only the last term of ℓ in Equation (1) depends on β_0, β_1
- Maximizing ℓ over $\beta_0, \beta_1 \iff$ minimizing $\sum_i (y_i - \beta_0 - \beta_1 x_i)^2$, which is the same as the OLS criterion

Question: what does this imply about $\hat{\beta}_{0,MLE}, \hat{\beta}_{1,MLE}$ versus the OLS estimates?

- They are identical — under the normal-error model, the the maximum likelihood estimates of β_0 and β_1 coincide exactly with the least-squares estimates

2 Estimating the Error Variance

MLE of σ^2

Differentiating ℓ (Equation (1)) with respect to σ^2 and solving:

$$\begin{aligned} \frac{\partial \ell}{\partial \sigma^2} &= -\frac{n}{2\sigma^2} + \frac{SSE}{2\sigma^4} = 0 \\ \hat{\sigma}_{MLE}^2 &= \frac{SSE}{n} \end{aligned} \quad (2)$$

- Compare to the OLS/unbiased estimator: $MSE = SSE/(n-2)$
- $\hat{\sigma}_{MLE}^2$ divides by n , not $n-2$ — it is *biased*: $E[\hat{\sigma}_{MLE}^2] = \frac{n-2}{n} \sigma^2$
- MSE is the bias-corrected version of $\hat{\sigma}_{MLE}^2$

The Difference, in Our Example

Rocket-propellant data: $n = 20$, $SSE = 166,254.65$

$$\begin{aligned}\hat{\sigma}_{MLE}^2 &= \frac{SSE}{n} = \frac{166,254.65}{20} = 8312.73 \\ MSE &= \frac{SSE}{n-2} = \frac{166,254.65}{18} = 9236.37\end{aligned}\quad \text{by Equation (2)}$$

- The MLE underestimates σ^2 by about 10% here — $n = 20$ is small enough for the bias to matter
- The two converge as n grows

Where MLE and OLS Diverge, in General

	OLS	MLE
Needs a distribution?	No (mean, variance only)	Yes (full distribution)
β_0, β_1 estimates	Least squares	Identical, under normality
σ^2 estimate	Unbiased, $SSE/(n-2)$	Biased, SSE/n
Framework	Geometric	Probabilistic
Optimality	Gauss–Markov (BLUE)	Asymptotic efficiency

3 Comparison

Statistical Advantages of MLE

- Asymptotically efficient: attains the Cramér–Rao lower bound as $n \rightarrow \infty$, under correct specification
- Provides a general toolbox: standard errors from the Fisher information, likelihood-ratio/Wald/score tests, AIC/BIC for model comparison
- Generalizes to models for which OLS has no natural analogue, for example: logistic regression, Poisson regression, and other generalized linear models (GLM)
- Consistent and asymptotically normal under mild regularity conditions

Where OLS Still Wins

- Gauss–Markov: OLS is BLUE assuming only zero-mean, constant-variance, uncorrelated errors — *no* distributional assumption required
- MLE’s efficiency and consistency rely on the assumed distribution being (at least approximately) correct
- Here, $\hat{\beta}_0, \hat{\beta}_1$ are unbiased in both approaches, but $\hat{\sigma}_{MLE}^2$ is biased while $\hat{\sigma}_{OLS}^2 = MSE$ is not